Forecasting Weekly German Electricity Demand Using Statistical, Machine-Learning and Neural Models.

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GitHub Link: https://github.com/saimahesh1810/electricity-demand-forecasting.git

1. Introduction:

Accurate electricity-demand forecasting supports power-system planning, scheduling and operational decision-making. Weekly electricity demand is influenced by annual seasonality, recent demand, temperature and public holidays, but increasing model complexity does not always lead to better forecasts.

This study compares several approaches for forecasting weekly German electricity demand: mean, naïve, seasonal-naïve and drift benchmarks, SARIMA, SARIMAX with weather and holiday covariates, a feature-based Gradient Boosting model and an LSTM neural network. The final 104 weeks were reserved as an unseen test period, and all models were evaluated using MAE, RMSE, MASE, Bias and sMAPE.

The main objective was to determine whether complex statistical, machine-learning or neural models provided a meaningful improvement over seasonal naïve. The analysis also examined data leakage, differencing choices, the contribution of temperature and holiday covariates, model interpretability and operational suitability. Attention was given to whether small numerical improvements justified greater complexity, uncertainty and maintenance requirements.

The analysis addresses the following main questions:

1. Do statistical, feature-based or neural models meaningfully outperform seasonal naïve?
2. Do temperature and holiday covariates improve forecast accuracy?
3. How do model complexity, interpretability and uncertainty affect the choice of an operational forecasting method?
4. Which model provides the best balance between accuracy and practical maintainability?

2. Data and Preprocessing:

The electricity-demand data were obtained from the **Open Power System Data (OPSD) Time Series package**. The analysis used the hourly single-index file **time_series_60min_singleindex.csv**, which contains harmonised electricity-system measurements for several European countries. German electricity demand was extracted from the column DE_load_actual_entsoe_transparency, while utc_timestamp was used as the time index. Only observations from 1 January 2015 onward were retained.

The original load values were reported in megawatts and were converted to gigawatts to improve readability. Invalid values were converted to missing observations, timestamps were sorted chronologically, and the hourly load series was grouped into consecutive weekly intervals. Each weekly observation represented the mean demand across a complete 168-hour period. Incomplete weeks were excluded so that all target values were calculated from equal-length periods. The resulting forecasting target, load_gw, therefore represented average weekly German electricity demand in gigawatts.

German public holidays were added as calendar covariates. Two weekly holiday features were constructed, **holiday_days**, which counted the number of public holidays within each week, and **has_holiday**, which indicated whether a week contained at least one public holiday. These variables were included because industrial, commercial and household consumption patterns may change during holiday periods.

After the electricity, weather and holiday data had been aggregated and aligned, the final processed dataset contained eight variables:

- load_gw
- temp_mean
- temp_min
- temp_max
- heating_degree_days
- cooling_degree_days
- holiday_days
- has_holiday

The final dataset contained 300 weekly observations from **1 January 2015 to 24 September 2020** with no missing values. The first 196 weeks were used for training, while the final 104 weeks were retained as an unseen test set. A chronological split was used to preserve the natural forecasting order and avoid future information influencing model development.

To prevent leakage, lag and rolling features were created only from earlier load values, with rolling statistics based on a one-week shift. During recursive forecasting, predicted values not actual test-period demand were used to construct later inputs.

Realised test-period weather was supplied to **SARIMAX**, Gradient Boosting and **LSTM**, so these results are conditional forecasts. Holiday dates were known in advance, but future temperatures were not; an operational system would therefore require weather forecasts instead of observed values.

Data Summary:

Item	Description
Electricity source	Open Power System Data Time Series package
Source file	time_series_60min_singleindex.csv
Demand field	DE_load_actual_entsoe_transparency
Timestamp field	utc_timestamp

Original frequency	Hourly
Forecast frequency	Weekly
Target variable	Average weekly German electricity demand, load_gw
Final period	January 2015–September 2020
Final observations	300 weeks
Training period	196 weeks
Test period	Final 104 weeks
Additional variables	Temperature, degree days and German holidays

*Table 1-**Data Summary Table***

3. Exploratory Data Analysis:

The weekly electricity-demand series showed a clear recurring annual pattern, with generally higher demand during colder months and lower demand during warmer periods. This seasonal structure supported the use of a 52-week seasonal period in the seasonal-naïve, SARIMA and SARIMAX models.

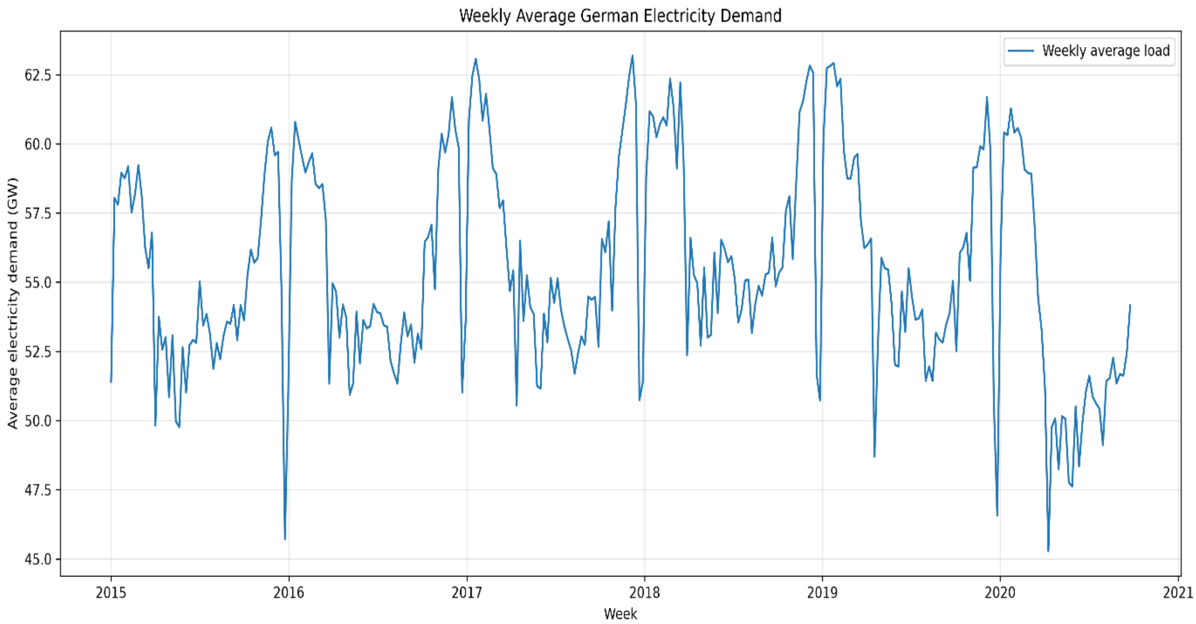

Figure 1 - Weekly average German electricity demand from January 2015 to September 2020.

The time-series plot also showed that demand did not follow an identical pattern every year. Several weeks contained sharper increases or reductions than the surrounding seasonal pattern, suggesting that holidays, weather conditions and irregular events may influence demand in addition to annual seasonality.

The yearly profiles provided further evidence of strong seasonal persistence.

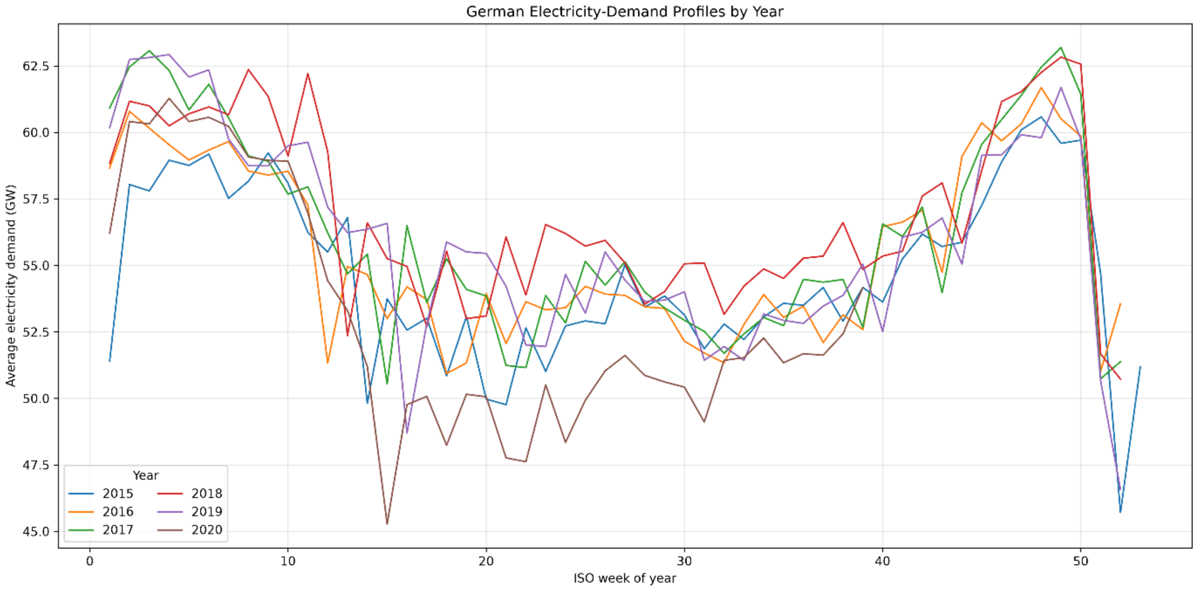

Figure 2-Weekly German electricity-demand profiles by year.

The similarity between corresponding weeks across different years explains why the seasonal-naïve benchmark was expected to be difficult to outperform. This was later supported by the Gradient Boosting feature importance results, where the 52-week lag was the dominant predictor.

Temperature also showed a visible relationship with electricity demand.

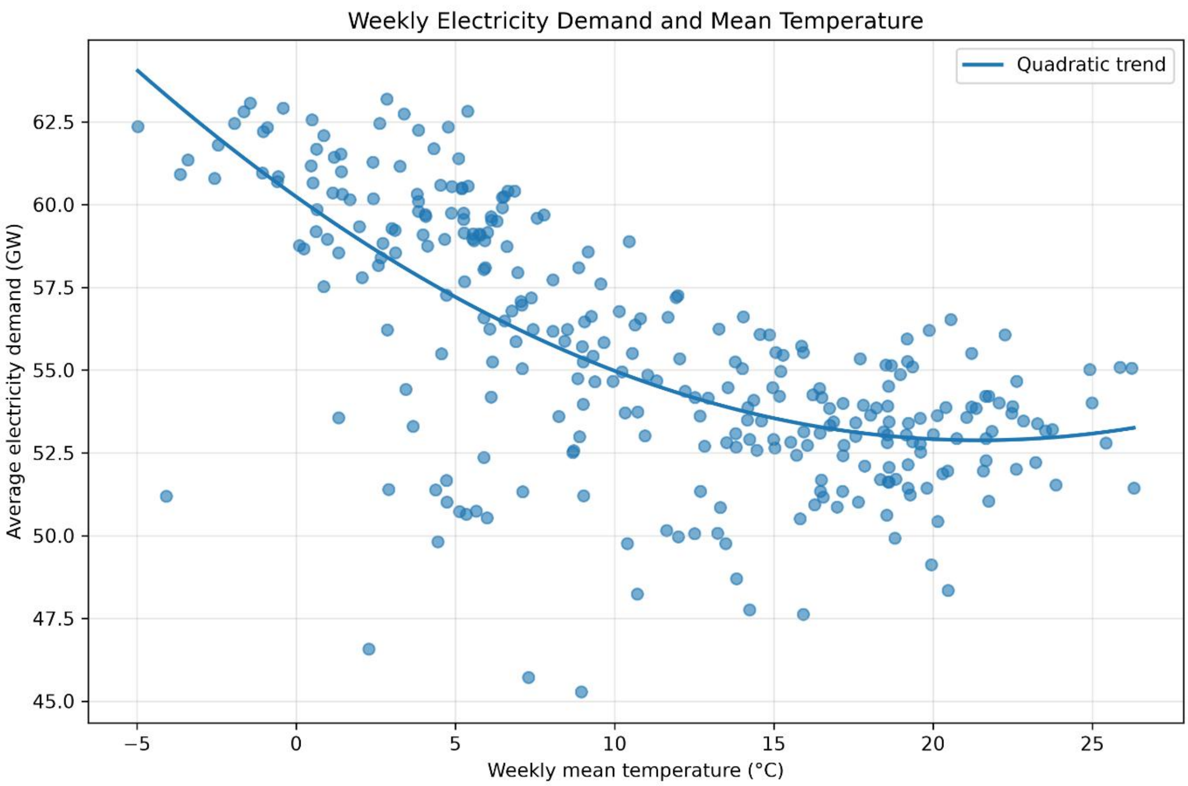

Figure 3-Relationship between weekly temperature and German electricity demand.

An Augmented Dickey–Fuller test was also applied to the training series. The original weekly demand series showed evidence of stationarity, although differenced versions were also examined during model development. Consequently, differencing orders were selected using a combination of stationarity evidence, model convergence, information criteria and forecast stability rather than relying on the statistical test alone.

Overall, the exploratory analysis identified three main forecasting signals:

1. strong annual persistence.
2. short-term dependence on recent demand.
3. additional effects associated with temperature and holidays.

4. Methodology and Evaluation:

The forecasting methods were selected to compare simple benchmarks, classical time-series models, feature-based machine learning and a neural network. All approaches used the same chronological split and were evaluated over the final 104 weeks.

4.1 Benchmark Models:

Four benchmarks were implemented: **mean, naive, seasonal naive and drift**. The seasonal-naïve model predicted demand using the corresponding week 52 weeks earlier and was treated as the principal benchmark because the exploratory analysis showed strong annual persistence.

4.2 Statistical models:

SARIMA was used to represent autoregressive, moving-average and seasonal dependence. Candidate non-seasonal orders were searched across combinations of p, $d$and q, with the seasonal period fixed at 52 weeks. Selection considered training AIC, convergence and numerical forecast stability. The final stable model was:

SARIMA (2,2,1) (1,0,1)52

SARIMAX extended this framework using temperature and holiday covariates. The final specification was:

SARIMAX (1,1,1) (1,1,1)52

The seasonal period of 52 represented yearly recurrence in weekly data. The non-seasonal and seasonal differences were used to model changes in level and annual seasonal differences. Candidate covariate sets were selected using training AIC, with mean temperature and holiday count providing the lowest value among stable models.

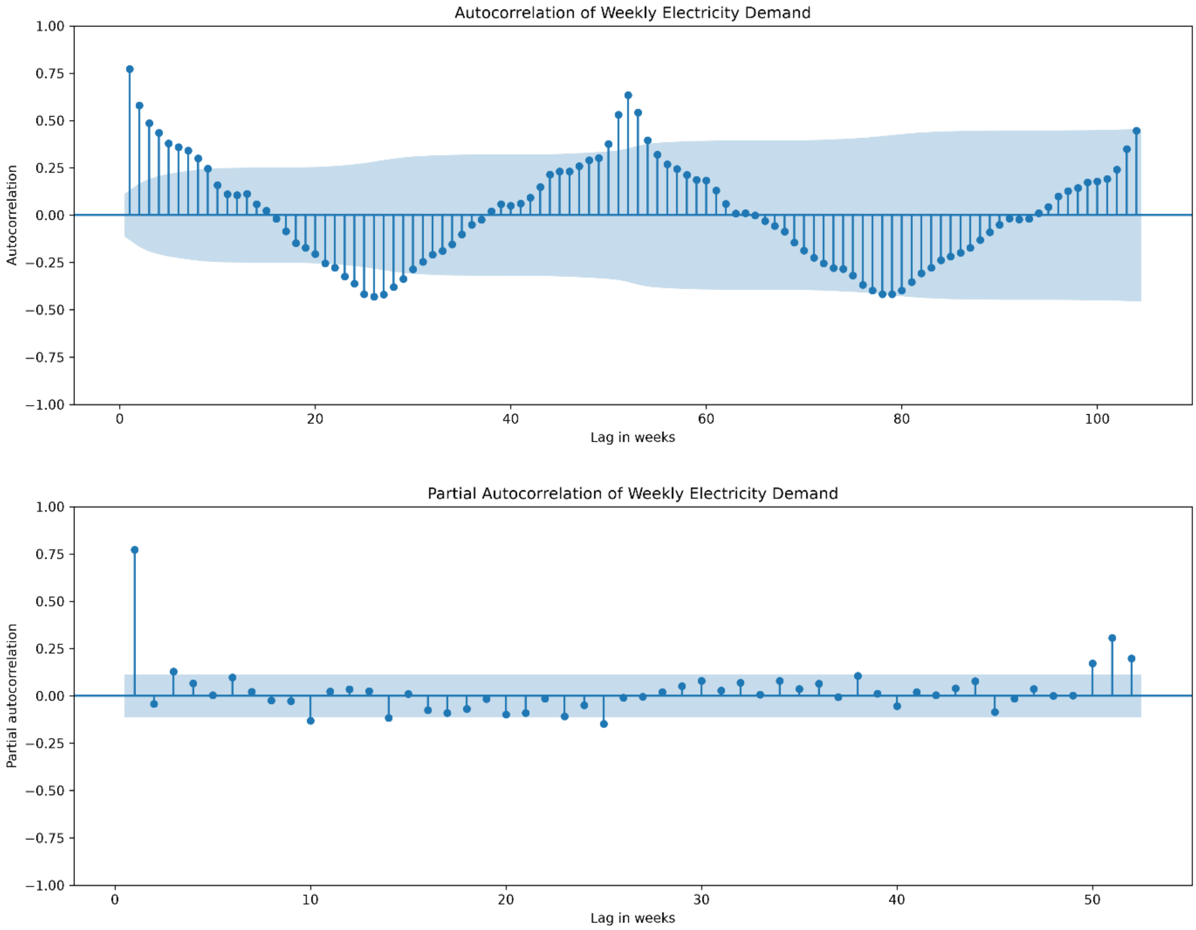

Figure 4-ACF and PACF of weekly demand. Persistent dependence supported autoregressive modelling, while annual recurrence justified a seasonal period of 52 weeks.

4.3 Feature-based Gradient Boosting:

Gradient Boosting used load lags, rolling statistics, weather, degree-day, holiday, cyclical calendar and trend features. The 52-week lag represented annual persistence, while shorter lags captured recent dependence. Rolling statistics were calculated from a one-week-shifted target, ensuring that the current load value was excluded.

Forecasts were generated recursively. After predicting one test week, that prediction was appended to the available history and used to construct features for the next week. Actual test-period demand was never supplied to later forecasts.

4.4 LSTM neural network:

The LSTM used a 52-week sequence containing load, mean temperature and holiday count. The model consisted of an LSTM layer, dropout and dense layers. Scaling parameters were estimated from training data only, and the final 26 training sequences were used for validation. Early stopping restored the weights from the lowest validation loss and reduced unnecessary overfitting.

The LSTM also used recursive fixed-origin forecasting, meaning that previous predictions replaced unavailable future load values in later input sequences.

4.5 Conditional covariates

Realised test-period temperature was supplied to SARIMAX, Gradient Boosting and LSTM. Therefore, their forecasts were conditional on observed weather. Holiday dates were known in advance, but temperatures over the full 104-week horizon were not. A live forecasting system would require meteorological forecasts or weather scenarios.

4.6 Evaluation metrics

Metric	Interpretation
MAE	Average absolute error in GW.
RMSE	Gives greater weight to large errors.
MASE	Error scaled against an in-sample seasonal benchmark.
Bias	Positive values indicate systematic over forecasting.
sMAPE	Symmetric percentage forecast error.

*Table 2-**List of Evaluation metrics.***

Model ranking was based primarily on MAE, with the other measures used to assess large errors, scale-independent accuracy and systematic bias. Model specifications were selected using training information rather than test-set accuracy.

5. Results:

All models were evaluated against the actual electricity demand observed during the final 104 weeks, from October 2018 to September 2020. This period was not used during model fitting and therefore represented genuinely unseen data collected after the training period.

5.1 Overall forecast accuracy:

Table 3 compares the models using MAE, RMSE, MASE, Bias and sMAPE. Lower values indicate better performance, except that Bias should ideally be close to zero.

Rank	Model	MAE (GW)	RMSE (GW)	MASE	Bias (GW)	sMAPE (%)
1	Gradient Boosting	2.3668	3.1334	1.6122	1.9051	4.3880
2	Seasonal naïve	2.4175	3.1549	1.6468	1.7720	4.4406
3	LSTM	2.7071	3.5358	1.8440	1.1743	4.9682
4	SARIMA	2.9901	3.6154	2.0368	1.1167	5.4633
5	Naïve	3.7319	4.3819	2.5421	−0.2931	6.7709
6	Mean	3.7761	4.4046	2.5722	0.5346	6.8505
7	Drift	4.0330	4.7181	2.7472	0.6361	7.3093
8	SARIMAX	4.2480	5.0295	2.8936	4.2371	7.5852

*Table 3-**Comparison of MAE and RMSE across all forecasting models.***

Gradient Boosting produced the lowest MAE and RMSE. However, its advantage over seasonal naïve was small. MAE decreased from 2.4175 GW to 2.3668 GW, an improvement of approximately 2.1%. RMSE improved by less than 1%. Therefore, the machine-learning model achieved the best numerical result, but not a large practical improvement.

The seasonal-naïve model substantially outperformed the mean, naïve and drift benchmarks. This confirmed that annual persistence was more informative than either a constant level or a simple extrapolated trend.

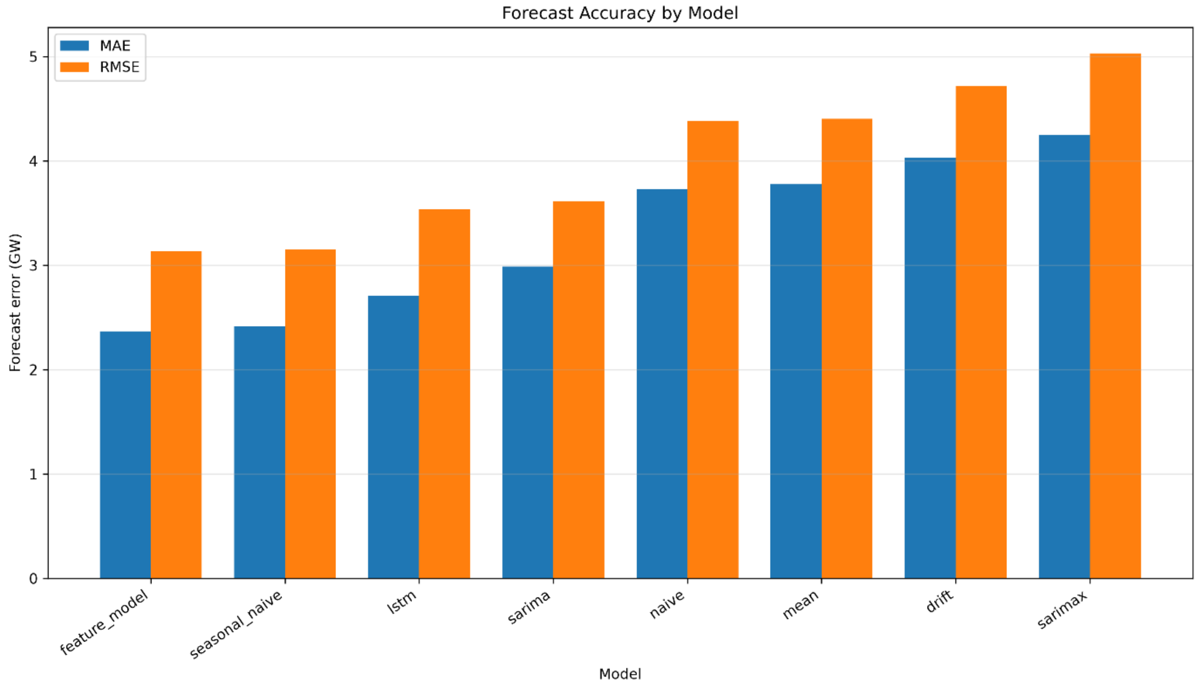

Figure 5-Comparison of MAE and RMSE across all forecasting models.

All reported MASE values were above one because MASE was scaled using an in-sample seasonal benchmark. The relative ranking remained consistent with MAE: Gradient Boosting ranked first, followed closely by seasonal naïve.

5.2 Benchmark forecasts:

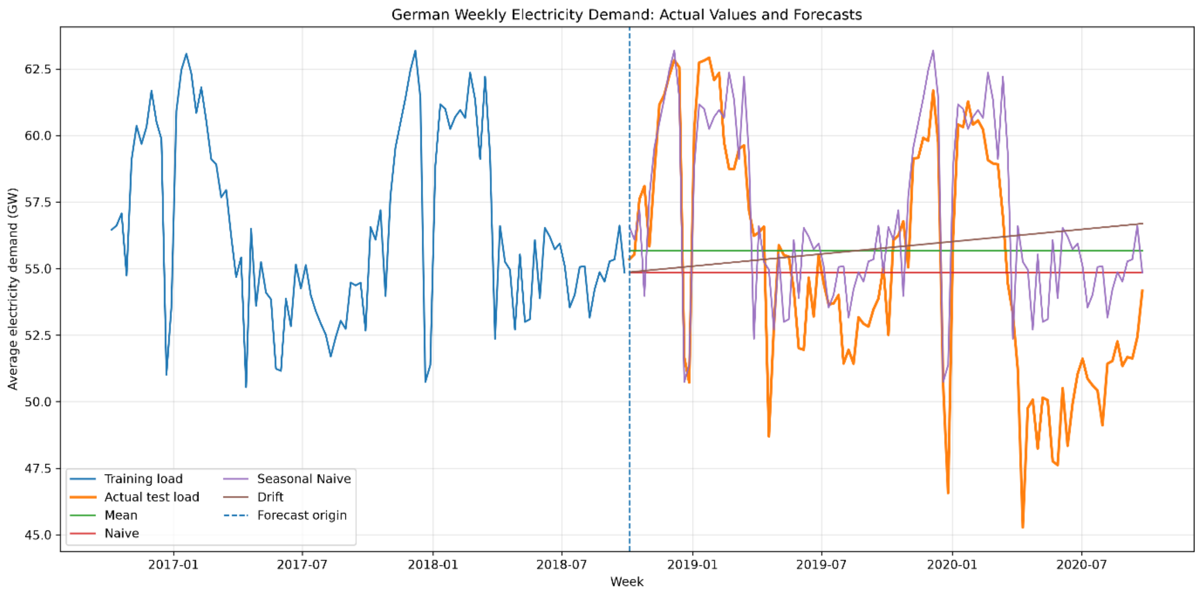

Figure 6-Mean, naïve, seasonal-naïve and drift forecasts compared with observed test-period demand.

The mean and naïve forecasts remained relatively constant and therefore failed to reproduce recurring seasonal increases and decreases. The drift forecast extended the historical trend but could not represent annual demand cycles, producing the weakest benchmark performance.

Seasonal naïve followed the observed annual shape more effectively because each forecast was based on demand from the corresponding week of the previous year. Its strong performance indicated that the demand profile was highly persistent across years.

5.3 SARIMA and SARIMAX forecasts:

The SARIMA model achieved an MAE of 2.9901 GW. It captured broad seasonal movement but was less accurate than seasonal naïve. Its positive Bias of 1.1167 GW showed moderate systematic over forecasting.

Ljung–Box tests for SARIMA residuals gave p-values of 0.0009 and 0.0026 at lags 12 and 26, indicating that shorter-term temporal structure remained unexplained. At lag 52, the p-value was 0.2557, suggesting that annual residual autocorrelation had largely been removed.

SARIMAX produced clean residual diagnostics, with Ljung–Box p-values above 0.05 at lags 12, 26 and 52. Nevertheless, its test accuracy was the worst among all models. Its MAE increased to 4.2480 GW, and Bias reached 4.2371 GW. Because Bias was almost equal to MAE, the model overforecast demand during nearly the entire test period.

5.4 Gradient Boosting forecast:

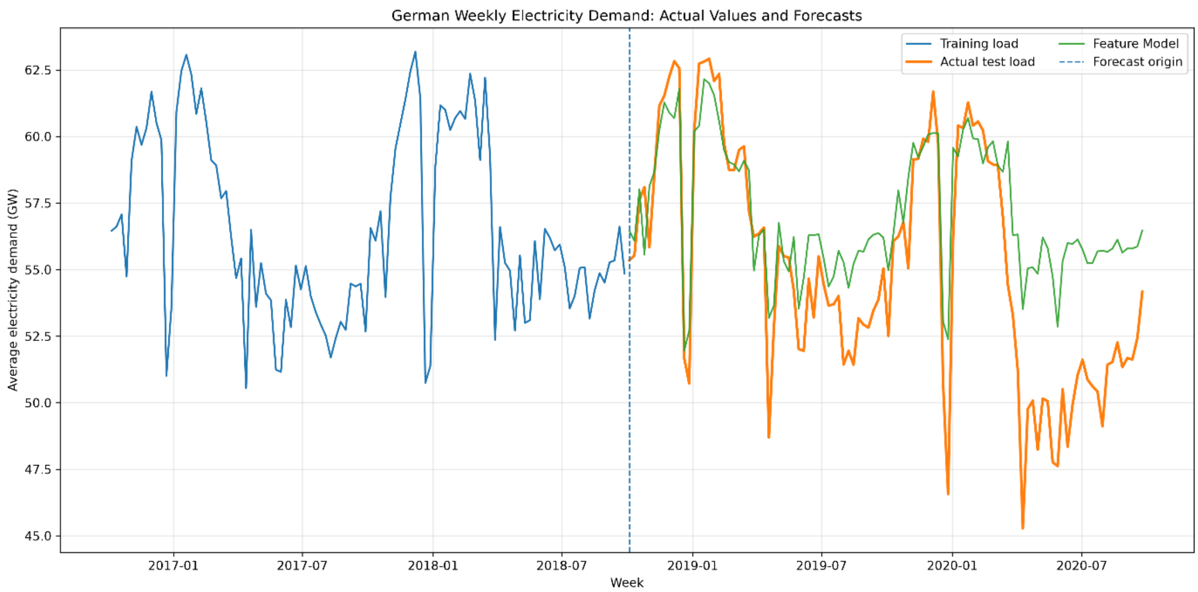

Figure 7-Recursive Gradient Boosting forecast compared with actual weekly demand.

The Gradient Boosting model achieved the best test accuracy:

- MAE: 2.3668 GW
- RMSE: 3.1334 GW
- MASE: 1.6122
- sMAPE: 4.3880%.

Its forecast range of 51.957 - 62.157 GW was plausible relative to the training range of 45.714 - 63.190 GW. The model followed annual demand variation more closely than the other complex methods, although it still over forecast on average by 1.9051 GW.

5.5 LSTM forecast:

The LSTM achieved an MAE of 2.7071 GW and RMSE of 3.5358 GW. It performed better than SARIMA and SARIMAX but did not outperform seasonal naïve or Gradient Boosting.

Its Bias of 1.1743 GW was lower than the Bias of both seasonal naïve and Gradient Boosting, indicating less systematic over forecasting. However, the LSTM predictions occupied a relatively narrow range of 54.507 - 60.704 GW. This compression suggests that recursive forecasts gradually moved toward the centre of the training distribution and did not reproduce some extreme peaks and troughs in the observed data.

5.6 Comparison with the observed test data:

The test-period observations showed that the annual seasonal pattern continued after the training period, which explains the strong performance of seasonal naïve and the importance of lag_52. However, the observed data also contained deviations from the previous year's profile.

Gradient Boosting captured some of these deviations using recent load, weather and holiday variables, producing the lowest overall errors. Nevertheless, the improvement was limited. SARIMA reproduced the general seasonal structure but missed shorter-term changes, while SARIMAX systematically predicted levels above the observed data. The LSTM generated smooth and plausible forecasts but underestimated the variability of the real series.

Overall, the comparison with the actual post-training observations showed that annual repetition was more reliable than model complexity. The best results came from either directly using the previous year's demand or making small nonlinear adjustments to it.

6. Critical Analysis: Assignment Questions:

- **Which models improved on seasonal naive?**
  Gradient Boosting performed best, reducing MAE from 2.4175 GW to 2.3668 GW, an improvement of about 2.1%. This was a small numerical gain rather than a clearly meaningful operational improvement. LSTM, SARIMA and SARIMAX all performed worse than seasonal naive.
- **How was data leakage avoided?**
  The split was chronological, preprocessing was fitted on training data only, and rolling features used shifted load values. During recursive forecasting, earlier predictions not actual test demand were used to create later inputs. Realised test-period temperature was used, so the weather-based models were conditional forecasts.

- **Why were the SARIMAX differencing orders and seasonal period chosen?**
  A seasonal period of **52 weeks** was used because the data were weekly and showed strong yearly recurrence. The final model used d=1 and D=1 remove local-level and annual seasonal changes. Although residual diagnostics were satisfactory, forecast accuracy remained poor.
- **Did temperature and holiday covariates improve accuracy?**
  No. SARIMAX had a higher MAE than both SARIMA and seasonal naïve and showed strong positive bias. Holiday dates were known at the forecast origin, but future temperatures were not and would require weather forecasts in practice.
- **How do SARIMAX, Gradient Boosting and LSTM compare?**
  SARIMAX was the most interpretable and provided native prediction intervals, but it was difficult to estimate and inaccurate. Gradient Boosting was the most accurate and offered useful feature importance, although it lacked native uncertainty intervals. LSTM was the least interpretable, required more maintenance and overfitted because of limited training data.
- **Which model is recommended for operational use?**
  Seasonal naive is recommended as the primary operational model, with Gradient Boosting as a challenger. Seasonal naïve was almost as accurate, simpler, transparent, inexpensive and independent of unknown future weather.

7. Future Improvement:

Future work should use rolling-origin evaluation to determine whether model rankings remain stable across multiple forecast periods. Realised test-period temperatures should be replaced by weather forecasts available at each forecast origin. National weather representation could also be improved by combining temperature data from several German regions rather than using Berlin alone. Longer historical or higher-frequency data may improve LSTM performance, while conformal or quantile methods could provide uncertainty intervals for Gradient Boosting and neural forecasts.

8. Conclusion:

The analysis showed that strong annual persistence dominated weekly German electricity demand. Gradient Boosting achieved the lowest forecast errors, but its MAE improvement over seasonal naive was only about 2.1%. LSTM and SARIMA added complexity without outperforming the seasonal benchmark, while SARIMAX produced clean residuals but poor and positively biased forecasts.

Seasonal naive is therefore recommended for primary operational use because it offers nearly the same accuracy as the best model while remaining transparent, inexpensive and independent of unknown future weather. Gradient Boosting should be retained as a challenger model and reassessed using longer datasets and rolling-origin evaluation.

9. References:

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